

its own data to these forms. Here are some data importing functions and their syntax in the form of examples.

An example for CSV and Excel files:

```
LOAD DATA LOCAL INFILE '/path/to/your/
csv/file/mydata.csv'

INTO TABLE database.table

FIELDS TERMINATED BY ',' ENCLOSED BY
''' LINES TERMINATED BY '\n';
```

An example for XML files:

```
LOAD DATA LOCAL INFILE '/path/to/your/
csv/file/mydata.xml'

INTO TABLE database.table

ROWS IDENTIFIED BY '<tag>'
```

It's obvious that for a database, a table has to be created before loading the data into it. Moreover, the table's structure should be compatible with the data that is to be loaded. For importing Excel data, the best way is to convert the Excel worksheet into a CSV file. This process filters out all the unprintable characters and meta data from the sample, making it ready for the SQL format.

Processing and filtering of data

This step is essential for further high-level analysis of data. Since SQL queries work on data of a particular format and the supporting data structures, filtering of data not only helps to eliminate unwanted errors but also to understand the data better. Among different data types, floating point numbers should be in proper format, and time-date data should be according to MySQL's YYYY-MM-DD format. Care is required to handle the null values also. There are a few SQL functions for identifying and subsequently cleaning the imported data. These functions are *LIKE()*, *TRIM()*, *REPLACE()*, *LEFT()*,

RIGHT(), *CONCAT()*, *UPPER()*, *LOWER()*, and *FORMAT()*. The most important SQL tools to handle data filtering dynamically are *COALESCE* and *CASE-END* statements. These have two forms — Search *CASE*, which is used in SQL query statements, and Simple *CASE*, which is used in stored programming.

In the cleaning process, the following needs to be taken care of:

- Formats of data
- Formats of decimal data
- Misspelt and mistyped data
- Null field values

Formats of imported data should match the SQL table format, and care should be taken while preparing data and the SQL tables. A NULL value field option can be incorporated into the SQL table structure itself. Spelling and other data arrangements can be handled with SQL functions like *TRIM()*, *LEFT()*, *RIGHT()*, *UPPER()*, *CONCAT()*, etc.

Higher-level dynamic filtering can be implemented using *COALESCE* and *CASE-END* statements. For instance, to filter a GENDER field with values like 'M', 'Male', 'F', 'Female', 0 and 1, *COALESCE-CASE* statements can be used to normalise the data as follows:

```
SELECT adm_no,stu_name,COALESCE(
    CASE WHEN sex="M" THEN "Male" ELSE
NULL END,
    CASE WHEN sex="Male" THEN "Male"
ELSE NULL END,
    CASE WHEN sex="F" THEN "Female"
ELSE NULL END,
    CASE WHEN sex="Female" THEN
"Female" ELSE NULL END,
    CASE WHEN sex="0" THEN "Male" ELSE
NULL END,
    CASE WHEN sex="1" THEN "Female"
ELSE NULL END
)
from STUDENTS
WHERE id >= 33 and id <= 42;

Adm_no Name Gender
```

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